

# Ryan Cho

## Senior Full Stack Software Engineer

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<https://github.com/Chomama> · New York, United States

### PROFESSIONAL SUMMARY

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Senior full stack engineer with 8+ years delivering secure, scalable web platforms in finance and enterprise environments. Deep experience across React and TypeScript UIs, .NET and Python services, PostgreSQL data design, and async processing with Redis and message queues. Recognized for end-to-end ownership, RBAC-centric architectures, audit-ready data practices, and collaborative delivery in remote agile teams.

### WORK EXPERIENCE

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#### Software Engineer

May/2022 - May/2026

##### Aspen Capital

Remote | 10th Floor, New York

- Architected **full stack features** across React and TypeScript clients and .NET services, owning delivery from API contracts through production rollout for capital-markets operations tooling.
- Designed **PostgreSQL schemas** and Entity Framework Core access patterns that became the source of truth for permissions, workflow state, and reporting across 15+ domain entities.
- Implemented **RBAC enforcement** at API and data layers, ensuring role-scoped queries and audit-friendly access patterns for internal and external users.
- Built **Redis caching** and RabbitMQ worker pipelines that offloaded document generation and reconciliation jobs, cutting median API latency by 35% during peak windows.
- Integrated **AWS S3 storage** for large artifacts with service-level encryption and lifecycle policies that kept web gateways lean and responsive.
- Shipped **SignalR real-time notifications** connecting portfolio dashboards to backend event streams used by 150+ daily operators.
- Authored **RFC-style technical proposals** aligning .NET service boundaries with Python analytics runners and shared data contracts across teams.
- Elevated engineering quality through **automated testing** guidelines, integration coverage for critical paths, and mentorship during code reviews across a 6-person squad.
- Partnered with product and design in **two-week agile releases**, converting complex customer workflows into iterative, production-ready full stack solutions.

#### Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

##### Summer Analyst,

##### BNP Paribas

Remote | Seventh Avenue, New York

- Engineered **C# and .NET services** supporting trade operations, regulatory exports, and risk dashboards used across New York markets infrastructure.
- Developed **immutable audit pipelines** with signed checkpoints and chain continuity checks, enabling evidence parity between C# services and Python validation tooling.
- Tuned **PostgreSQL transactional writes** using advisory-lock coordination for high-concurrency audit ingestion without sacrificing consistency.
- Delivered **React and TypeScript interfaces** and REST APIs that surfaced compliance datasets with role-aware filtering for analysts and operations teams.
- Introduced **Redis-backed caching** for session state and reference data, stabilizing response times through intraday trading volume spikes.
- Supported **AWS KMS envelope encryption** patterns for sensitive payloads at rest and in transit across multi-environment deployments.
- Collaborated with security stakeholders on **access control reviews**, retention policies, and control implementations aligned with enterprise compliance frameworks.

- Maintained **CI-driven automated testing** and release workflows that improved confidence for production deployments in regulated financial systems.

## Software Engineering Intern

Jun/2018 - May/2019

### Chexology

Remote | 17th Floor, New York

- Contributed **React and TypeScript UI** for compliance checklist workflows, improving task completion clarity for restaurant field teams.
- Built **C# REST endpoints** powering authentication, document uploads, and status transitions within a document-management platform.
- Implemented form validation and API integration patterns that reduced invalid submission retries during pilot rollout.
- Added unit tests and participated in **agile sprint planning**, standups, and peer reviews with senior engineers.
- Optimized client rendering and request batching to improve responsiveness on mobile connections in production pilots.
- Documented API usage and component behavior to accelerate onboarding for subsequent engineering contributors.

## Web Developer

Sep/2017 - Nov/2017

### Bayesquare Foundation

Remote | New York

- Developed **responsive web applications** for nonprofit outreach campaigns using component-based JavaScript and modern CSS layouts.
- Integrated external APIs for donor management and event scheduling within a **remote project cadence** and tight delivery timelines.
- Built accessible form experiences and navigation flows that supported fundraising staff across desktop and mobile browsers.
- Translated stakeholder feedback into iterative UI improvements released through version-controlled deployment workflows.
- Maintained clean Git history and lightweight deployment scripts enabling dependable site updates for campaign launches.

## EDUCATION

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### New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

## SKILLS

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**Frontend:** React, TypeScript, JavaScript, Vite, Tailwind CSS, SignalR, HTML/CSS, Responsive UI

**Backend:** C#, .NET, .NET Core, Minimal APIs, Entity Framework Core, Python, REST APIs, RBAC

**Databases:** PostgreSQL, Redis, SQL Server, Entity Framework, Schema Design, Transactional Auditing

**Cloud & DevOps:** AWS (S3, KMS), Docker, CI/CD, GitHub Actions, Envelope Encryption, Infrastructure as Code

**Monitoring & Tools:** Automated Testing, xUnit, Jest, Git, GitHub Copilot, Application Logging, Performance Profiling

**Workflow & PM:** Agile/Scrum, RFCs, Code Reviews, Technical Direction, Cross-functional Collaboration, Remote Delivery